

Empowering Children for Social Justice–Design Research in the Context of Children's Computing Education

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Empowering Children for Social Justice–Design Research in the Context of Children's Computing Education

Netta Iivari(✉), Leena Ventä-Olkkonen, Heidi Hartikainen, Sumita Sharma, Essi Lehto, and Jenni Holappa

INTERACT Research Unit, University of Oulu, 90014 Oulu, Finland
{netta.iivari,leena.venta-olkkonen,heidi.hartikainen,
sumita.sharma,essi.lehto,jenni.holappa}@oulu.fi

Abstract. In the contemporary digitalized society, it is pivotal to provide children with skills to ideate, design and develop digital technology and to critically reflect on it and its consequences. In the current world full of crises, it is also crucial to make visible for children that digital technology is never value-free and that it can be used for making the world a better place but also for discrimination, marginalization, and oppression. Information Systems (IS) scholars have not addressed these topics with children, while IS as a discipline would provide very suitable expertise for this. We, inspired by frameworks on empowerment and social justice, have invited children to tackle a wicked societal problem of bullying through design of digital technology as part of their computing education, and while doing so, aimed at empowering children to empower those suffering from bullying, i.e., advocating social justice. In this design research study, we analyze and showcase our empowering design process and outcomes. We contribute to IS literature by offering insights on how IS research can contribute to children's empowerment and social justice oriented computing education. These insights serve also IS researchers working with adults and pursuing social justice and empowerment agendas.

Keywords: Children · Empowerment · Social Justice · Computing Education

1 Introduction

There is an increasing pressure set for the computing education of children of today: such education should not only equip children with skills and competences to make meaningful use of digital technologies, but such education should encourage children to start shaping and making such technologies as well as approaching them critically (e.g., [1–10]). Overall, children should be educated to make informed decisions on digital technologies concerning their future everyday life and society [1].

Information Systems (IS) literature has remained quite silent about this important topic. There are few IS studies highlighting children as an important user group [11–13]

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and few studies arguing we should consider also children's computing education, not only that of adults [14–16]. Some IS studies have concentrated on student marketing and recruitment for senior highschoolers, addressing their computing education (e.g.[16, 17])– but even these studies are rare as IS education is oftentimes targeting students already in higher education in their marketing and recruitment efforts [16, 17]. We argue that IS research should direct its gaze towards children. Overall, we feel there is a lot IS research could offer to children's computing education.

Among such issues is the critical research tradition and the critical approach towards societal consequences of digital technology. Those are becoming ever more significant, also for the younger generation and their computing education. We are currently living in a turbulent world, suffering from crises of several kinds, feeling anxiety regarding our own future as well as the future of humankind and the globe. We feel that addressing these global crises and associated collective anxiety should be on the agenda of computing education, too. Indeed, recent research on children's computing education has started integrating critical and social justice topics into the education [2, 3, 6–10, 18, 19]. This literature base argues that children should be empowered to address important societal problems relating, e.g., to oppression, inequality, injustice, marginalization or climate change through design of digital technology (for a review, see e.g., [20]). However, currently children's computing education, although being globally developed under the labels, e.g., of Science, Technology, Engineering and Mathematics, computer science, programming, computational thinking, or digital fabrication and making education, remains too technology focused and exclusive [1–10]. IS research, being inspired by the critical research tradition and having a strong background in approaching digital technologies and their consequences critically already for decades (e.g., [21–24]), offers interesting avenues for cross-fertilizing the emerging stream of research on children's critical and social justice oriented computing education.

We have experimented with children's computing education that critically addresses important societal problems through design of digital technology, contributing to the currently very limited knowledge base on children's computing education of this type. Particularly we have addressed together with children the problem of bullying, which is a wicked–serious, complex, and extremely difficult to solve–problem of the current society [25, 26]. Bullying refers to aggressive behavior deliberately and repeatedly causing harm to a peer by someone in a more powerful position [25]. In this study we report on a project in which we, inspired by the frameworks on empowerment and social justice, have invited children to tackle bullying and while doing so, aimed at empowering children to empower those suffering from bullying, i.e., we have invited children to advocate social justice. As our research question we ask: How can critical perspectives, particularly on empowerment and social justice, be introduced into children's computing education curriculum? For answering the question, we will scrutinize the design process and its design outcomes from the viewpoint of empowerment. Our design research study contributes to IS literature by offering yet missing but badly needed insights on how IS research can contribute to children's computing education with empowerment and social justice lenses. Our insights serve IS researchers interested in children's computing education but also those working with adults and pursuing social justice and empowerment agendas.

2 Social Justice and Empowerment Lenses

Social justice is a complex concept with many possible interpretations and traditions [18, 19, 27, 28]. Topics such as racism, sexism, and classism are addressed under this umbrella term [27]. A large number of IS studies can implicitly be seen to address the topic, but there are also some studies that explicitly address social justice. These studies have considered how digital technologies contribute to social justice or injustice [28–30] as well as how critical IS research can aim towards social justice [23, 31].

Fraser [32, 33] offers a widely cited framework to approach social justice, utilized also in IS (e.g., [28, 29]). According to her, social justice can be approached from three perspectives: resource redistribution, cultural recognition and political representation. According to her [32, 33] all are needed for social justice to happen. She advocates the norm of participatory parity, relating to which she maintains that social justice "requires social arrangements that permit all (adult) members of society to interact with one another as peers. For participatory parity to be possible, I claim, at least two conditions must be satisfied. First, the distribution of material resources must be such as to ensure participants' independence and "voice." Second, the institutionalized cultural patterns of interpretation and evaluation express equal respect for all participants and ensure equal opportunity for achieving social esteem. Both these conditions are necessary for participatory parity. Neither alone is sufficient." [33] Thus, her view focuses on the economic structure but also on the culturally defined status hierarchies of society. Moreover, she [32] adds political dimension of representation to her theory of justice, underscoring the questions of who are included, who are excluded, who has a voice: "If representation is the defining issue of the political, then the characteristic political injustice is misrepresentation. Misrepresentation occurs when political boundaries and/or decision rules function to wrongly deny some people the possibility of participating on a par with others in social interaction–including, but not only, in political arenas". Fraser [32] discusses misrepresentation and misframing, the former referring to those included being wrongly denied from participation while the latter referring to exclusion of some from participation at all in the contests over justice.

In the case of bullying and children, we interpret bullying as an example of social injustice and see that tackling of it is more of a question of cultural recognition and political representation than redistribution of material resources, at least in our context. We interpret the tackling of bullying to concern "equal respect for all participants and (…) equal opportunity for achieving social esteem" [33]. We also see there are issues relating to the political dimension at play, addressing the boundaries and rules over participation and decision-making: who are included, who are excluded, who are allowed to have voice (see [32]), among children. In our case, we invite children to tackle injustice with the means of design of digital technology.

In our work we derive inspiration from the literature addressing empowerment. In our case, we aim to empower children to empower their peers. Empowerment is a complex concept with many disciplines and traditions, most notably psychology and organizational and management studies, but has also been addressed in IS research (see e.g. [34]). We approach empowerment with three distinctions: mainstream vs. critical, individual vs. collective, and motivational vs. relational (e.g. [34–41]). Considering the

mainstream vs. critical distinction, within the former empowerment is seen to entail giving people some more power of decision for the purpose of motivating them to perform better, while within the latter this view is rejected and it is maintained that empowerment instead necessitates that the oppressed question the oppressing status quo and combat the oppression, in this way liberating themselves as well as the oppressors [34, 36, 37]. In the context of bullying and children, the former view maintains that the bullied should be provided some more power of decision to help to prevent or reduce bullying while the latter view necessitates that the bullied question the state of the affairs and liberate themselves and the bullies from bullying. Considering the individual vs. collective distinction, within the former empowerment is seen to concern only individuals, while within the latter, it is seen to concern entire collectives, empowerment of which and feeling of social responsibility among all are highlighted [37–39]. In the context of bullying and children, the former concentrates only on the bullied and bullies, while the latter sees that entire communities should be involved, approaching bullying as a collective problem requiring collective action taking. Finally, considering the relational vs. motivational distinction, within the former empowerment is seen as power over someone or something [35, 36], while within the latter, empowerment is seen as increased task motivation, which is strengthened by perceived choice, meaningfulness, impact, and self-efficacy [36, 40]. In the context of bullying and children, the former sees that those bullied should gain more power over the bullies, while the latter underscores everyone's increased motivation to prevent or reduce bullying, which is seen to be increased by perceived choice, meaningfulness, impact and self-efficacy.

3 Methods

Methodologically, this study represents design research. In IS, Design Science Research (DSR) has become very popular during the past decades (e.g., [42–45]), DSR in IS research connects heavily with the design science model proposed by Cross [46]. DSR can be characterized by a desire to scientize design [46]. Cross [46], instead, argues for viewing design as a discipline in itself: not trying to fit it into the scientific method, but instead aiming to understand it as a rigorous culture of its own.

Along these lines, this study carries out design research, which aims at changing the world through design [47] and it is about "proliferation of new realities" [48]. Such research generates design artefacts as outcomes, but also knowledge contributions [47, 48], for which characteristic is that the discipline itself is generative: it is changing the context in which it operates [48]. Therefore, the knowledge contributions also need to be seen as generative and suggestive [48]. Valuable is to place the interest on the design artefacts that are capturing and embedding our implicit theories [48].

This design research project has lasted for four years with the goal to explore critical design and critical making together with children. City of Oulu representatives suggested exploring critical design and critical making to tackle the problem of bullying at school. A steering group with representatives from the City of Oulu and youth research (with expertise on educational administration, computing education, antibullying interventions) was established. During the timespan, numerous design research interventions have been carried out with different classes of children and their teachers, ranging from

eight years old children to 15 years old. The steering group suggested the schools to work with. The different interventions have had somewhat different foci over the years as well as many practical factors shaping them. The design research project aimed to explore different approaches for critical design and critical making, due to which the approaches and methods have been changing and evolving over the years. The project aimed at exploring the approaches with different age groups, due to which the practical activities, task assignments and presentations had to be modified to meet the needs of the different age groups. Teachers were significant collaborators in this work. Finally, the project was executed in the middle of the pandemic, which heavily influenced how the work was carried out with different groups of children.

This analysis focuses on three interventions, in which participants envisioned future technological solutions to prevent or reduce bullying at school. In interventions A and B, the participants, relying on design fiction [49] and theater of the oppressed approach [50], created low-fi prototypes of their imagined solutions that they incorporated into theater performances to showcase their use and impact for their school community, while in intervention C the participants, relying on design activism approach [51] and building on the design ideas of the previous interventions to advocate for genuine impact for the children's design in the realm of digital technology development [52], conducted activism campaigns to raise awareness about the issues of bullying and the impact of their proposed solutions in their school community. Intervention A was conducted with a 2nd grade class in Fall 2020, intervention B with a 6th grade class in the academic year 2020–2021, and intervention C with a 7th grade class in Spring 2022. Before the study, information about the research, as well as research consent and assent forms were distributed to the students and their guardians. All students took part in the sessions as part of their schoolwork, but research data was only included from the students who had parental consent and gave their assent to research. Participant information is presented in Table 1.

Table 1. Participant information (data shown from those we had consent)

	Grade level	Number of participants	Age of participants	Number of sessions
A	Second	25	7–9	18–45 min
B	Sixth	23	11–13	10–90 min
C	Seventh	38	13–15	6–45 min

The data collected from the interventions includes various kinds of participant generated ideation, design, prototyping and evaluation data as well as pre- and post-interviews or questionnaires. The data also includes video recording of specific sessions as well as different kinds of process data (field notes, photos, memos, project management data). The first four authors took the main responsibility for the data analysis. The analysis process was iterative and required a lot of discussion and negotiation among the analysts. The analysis entailed critically scrutinizing the design artefacts generated during this design research project by adult researchers and participating children. We focused on design artefacts addressing the design process and the design outcomes: for the former,

we examined the methodological material created by us for guiding the design process, while for the latter, we inquired the prototypes created by the children. We acknowledged that various theories have been embedded in and are underlying and shaping these design artefacts, but more strongly those created by us, researchers, than those created by children. First, we analyzed our design process, which entailed a lot of local variation, but also a set of general features, and abstracted an outline of the design process. Afterwards, we explicated the theories on empowerment underlying the design process. The next section presents the various empowerment theories embedded within our design process. Thereafter, we paid attention to the design outcomes, while in this case instead of abstracting a general outline or locating the most representative cases, we intentionally searched for variety within: we identified children's prototypes that showcase different empowerment theories as featuring in them.

4 Research Ethics

Considering that the project centers around the delicate matter of bullying, which can result in significant and enduring psychological consequences for those involved, we approached this subject matter with great care. This was particularly important when dealing with children. In order to maintain the utmost ethical integrity, we fostered a close partnership with the teachers. Prior to activities with the students, we engaged in an interview with their teacher. This interview encompassed various subjects, including the topic of bullying within the class and the teacher's and school's established protocols and strategies for addressing bullying. The interview served to confirm that the class did not have significant problems with bullying and that the school had effective procedures in place to address bullying incidents when they did occur. In addition, we collaborated with the teacher prior to every activity to ascertain their suitability for the students. Following each session, we held discussions with the teacher to collect their input. Throughout all activities, the teacher was actively involved in offering guidance and support. Lastly, the teacher prearranged groups in a way that balanced the students' strengths, thereby reducing the potential for discrimination or bullying. Throughout the project, we maintained a keen awareness of the power dynamics within the classroom and took deliberate actions to ensure that participation was fair and just. We were committed to never pressuring children to share their thoughts and we stressed the importance of grounding the outcomes in imagination rather than reality.

5 Design Process

Figure 1 summarizes our design process from the viewpoint of empowerment theories integrated into it, aiming at empowering children in relation to 1) bullying and 2) design of digital technology, to strive for social justice.

We maintain that all the empowerment theories identified in the literature are featuring in the design process. We started by sensitizing the children with the problem of bullying (with scenario and persona creation and letter writing) and with the design of digital technology (with start interviews/questionnaires, creative reflection tasks on own

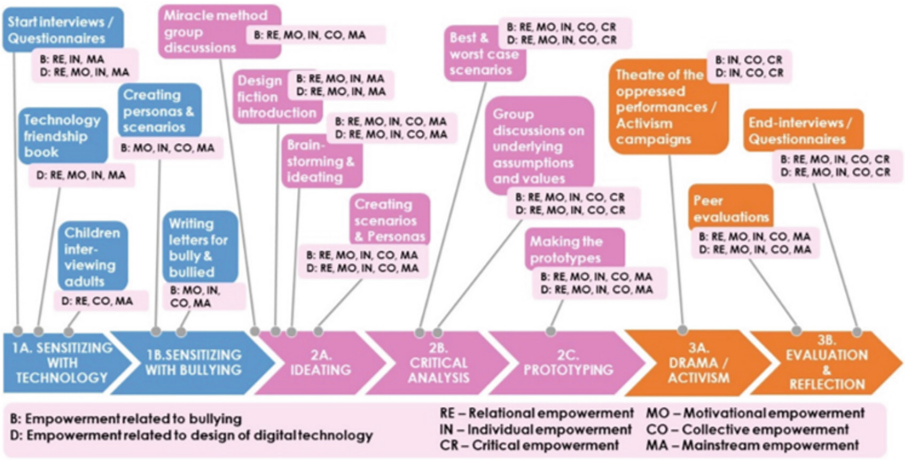


Fig. 1. Empowerment theories embedded in the design process.

relationship with digital technology and its design): making those visible for the children and inquiring their stances towards them, including their associated motivation and skills. We tried to make visible early on that children can make a difference regarding both bullying and digital technology (**relational empowerment**): they do not have to be accepted as they are, but instead one can take action and influence them. We also aimed at motivating children to work against bullying as well as at utilizing design of digital technology in that quest (**motivational empowerment**): the sensitizing tasks relating to bullying aimed at building empathy to motivate the children. The design process also aimed at developing children's competences as well as integrated choice and impact–we left many aspects open for children to decide and have an impact on.

We started by approaching the children as individuals (**individual empowerment**), but we soon introduced the idea of bullying as a collective problem through miracle method discussion on a world where a miracle had happened, and bullying had disappeared–addressing it from the perspective of different stakeholders. We asked the children to consider both bullying (with personas, scenarios and letter writing) and digital technology (with interview of an adult) from other persons' perspective, to design for solutions that address bullying as a collective problem (with brainstorming, design fiction, persona and scenario creation, prototyping), to evaluate those designs from other persons' perspectives (with peer evaluation) as well as to try to mobilize their broader school community for action taking against bullying (with theater of the oppressed performances, activism campaigns) (**collective empowerment**).

In the beginning, we approached empowerment in quite a mainstream sense: giving children some more power of decision regarding design of digital technology and tackling of bullying and motivating them for better performance (**mainstream empowerment**), while during the process we integrated critical concerns more and more strongly into the process: already during the sensitizing phase with personas, scenarios and letter writing, we invited children to critically analyze the status quo from the viewpoint of the bullying

problem and the bullied suffering from oppression. Later on, we encouraged the children to critically analyze their own designs from the viewpoint of best- and worst-case consequences, values and underlying assumptions. Thereafter, we asked the children to build theater of the oppressed performances or activism campaigns to mobilize broader collectives to tackle the oppression and to strive towards social justice. In the end interviews/questionnaires, we also asked children to critically evaluate and reflect on both the process and the outcomes, explicitly also discussing the goal of empowerment of the oppressed with them (**critical empowerment**).

6 Design Outcomes

The children's design outcomes are shown in Fig. 2.



Fig. 2. Empowerment theories embedded in the design outcomes.

A *super-girl to super-truck* (SGST) transformer was envisioned by a group of 2nd graders. The super-truck form of the design enables the super-girl to reach the bullying situation to address bullying as and when it occurs - by intervening and making the bully apologize using mind-control. The fidget spinner aids the super-girl in mind control, while the fire blades enable the super-truck to travel faster. This act of stopping the bullies and making them apologize includes the forms of **individual** and **relational empowerment** where the power dynamics between the bully and the bullied is changed and there is increased power over the bully, who is made to apologize. In this case the power is held by technology. This prototype also indicates a bit towards **critical empowerment** in the sense of questioning and combatting the oppressing status quo in the sense of mindset of the bully. However, we acknowledge that the idea of mind control poses many ethical challenges that do not align with the critical empowerment ideals.

The *PFB (Prevention for Bullying) robot* was designed by a group of 2nd graders. The robot is a friendly figure for preventing bullying (in the future) by motivating the bully

to change through reflection and discussions, asking them why they bully. We interpret the forms of **individual** and **motivational empowerment** to feature in this prototype: the bully is motivated to stop bullying, addressing the underlying issues through critical reflection, and to become friends with the person who was being bullied. Motivating the bully to change and become friendly can also be seen as a form of **critical empowerment** in the sense of radically changing the current state of affairs, more specifically the mindset of the bullies. In the actual bullying situation, the robot distracts the bully, moving their attention to the robot from the person being bullied to stop the bullying, another way **motivational empowerment** featuring in the design. The automated monitoring system to detect bullying enables the robot to arrive at the bullying scene, without human effort, to address the situation.

Magic headband was ideated by a 6th graders group. Magic headband's main idea was to change the world to a better place for us all by making bullying disappear, which can be seen to represent forms of **collective** and **critical empowerment**. The headband detects bullying, racism, and other kind of discrimination nearby and affects people's minds by changing bullies into nice people. The idea raises the same ethical questions than the first one. This idea clearly features **collective empowerment** as well: it is not concentrated on individuals but creates a positive atmosphere for all.

A tiger warrior robot was ideated by 6th graders. A tiger warrior robot monitors the school with its camera eyes, thus providing **collective empowerment** for the whole school community. It detects verbal bullying and sets punishments, but also rewards for good behaviour. We interpret that the idea engenders **relational empowerment** as it intervenes in bullying events and punishes the bullies, i.e., it introduces power over the bullies, while the punishment is seen to benefit the collective. In addition, the idea contains features of **motivational empowerment** by rewarding those who behave well.

BullyCounter mobile app was created by a group of 7th graders. Bullycounter can be seen to focus on **collective empowerment:** the users are children from all over the world who struggle with bullying and who can use the application to share their experiences and advice, helping each other out. The app can also be seen to include **individual empowerment:** while sharing experiences with others, children also receive help and insights to improve their own situation. If they require more help to solve the situation, they can choose to contact an adult. In addition, there are resources for self-help for those not comfortable discussing their situation on the platform–There they can find e.g., age-appropriate informational resources and helpline numbers. Users can also post to Bullycounter to raise awareness about the problem of bullying in general, which can be seen as a form of **critical empowerment.** The functionality to shame those who have been bullying others entails **relational empowerment** to those that have been affected by bullying, over the bullies.

7 Concluding Discussion

We reported on a design research project that engaged children to tackle a wicked societal problem of bullying through design of digital technology, inspired by theories on empowerment and social justice. IS research has so far remained ignorant of children's computing education, while having very suitable expertise to be utilized in that, especially in such that is having critical, empowerment and social justice orientation.

In this study, we analyzed and presented our design process and design outcomes with children, revealing how we–adult researchers and participating children–have integrated a variety of empowerment theories into the process and the outcomes. We pointed out that empowerment is a complex concept with multiple traditions and definitions. Through our analysis, we illustrated there are multiple ways by which one may address empowerment as part of the design process and design outcomes. We also pointed out that not all these ways are equally desirable. We underscore the value of collective and critical forms of empowerment (see [34, 36–39]), but we acknowledged all forms of empowerment featuring in the process and outcomes.

Our study also emphasized that social justice can be approached from multiple angles. Our empirical analysis did not yield equally rich insights into how social justice was embedded into and featuring in the design process and outcomes, while social justice acted as an important goal driving the entire project, particularly approached from the viewpoints of cultural recognition and political representation [32, 33]. The children's prototype descriptions were not detailed enough on this matter to decipher in detail which forms of injustice were particularly tackled. Our design process also remained on quite a general level regarding this matter. The design process documentation generated by children when sensitizing with the problem of bullying and their theater performances could provide more insights on this aspect and those will be analyzed in our future studies.

Our insights contribute to IS literature in several ways: we offer insights into how IS research can contribute to children's empowerment and social justice-oriented computing education as well as insights for critical IS research pursuing social justice and empowerment agendas. IS research has been inspired by critical research tradition for ages (see e.g., [21–24]). Adhering to the principles of critical IS research, we utilized concepts from the critical research tradition, were explicit about our value basis as well as asked children to do so and challenged the oppressive status quo and strived towards empowerment of individuals as well as collectives together with children (see [24]). In agreement with the prior IS literature, we have underscored socially responsible design and design research, critical design and design research, and the value of integrating criticality into those (see e.g., [22–24]). Our approach represents a participatory approach instead of an expert driven one (e.g., [22])–we invited local stakeholders, in our case children, to critically scrutinize the oppressing conditions of the status quo and to strive towards radical change.

Our approach enables renewing children's computing education. Even if critical, empowering and justice oriented aspects have already received some attention [1–3, 6–10, 18], these studies represent a small minority in the literature base and are very early attempts to introduce these concerns into children's computing education. Although this analysis mainly explicates the theories embedded in our design artefacts, our analysis also provides some evidence on the value of this work: our analysis showed that the children took the empowerment and social justice concerns seriously and took action for a better, more socially just, world. We already have some empirical evidence that our design process empowered the participating children (reported, e.g., in [49, 53]), indicating there is a lot of value in integrating this type of approach into children's computing education. However, we lack studies on whether the design outcomes are

really empowering other children, even if some empowerment theories were seen as picturing in them. Some of the design outcomes can even be seen as disempowering and even for the potentially empowering ones, future empirical studies are needed to understand whether they are actually empowering children.

As for the applicability of the findings in other contexts, we point out that these results have been gained in a particular country and with particular age groups, with many context specific aspects to be noted, such as the educational system. In different countries with different educational systems or with different age groups, changes may be needed for the approach. This is certainly the case for the individual activities, task assignments and materials used–even we had to modify them for the different age groups. Then again, the basic underlying approach worked well for all age groups. We actually think that this type of computing education approach could work also with adults, when properly modified. We claim that in IS education in universities, there is a lot to be learned from the approach discussed in this study. Critical IS scholars could utilize this approach in their educational activities and develop IS education to be more socially responsible, critical, empowering and justice oriented. This would enable their critical work to have a potentially wide-ranging real-world impact. We think our approach could also be used outside the context of education. IS researchers working with vulnerable or marginalized stakeholders or communities or with topics relating to equality and social justice may benefit from our approach to design and design research. By following our approach, IS researchers could integrate critical research tradition inspired theories into their design research processes as well as invite their participants to rely on those theories in their design.

In addition to the limitations and paths for future work already discussed, there are others worth mentioning. We provided deeper insights into how empowerment theories featured in our design process and design outcomes than into how social justice theories did so. In the future, more in-depth analysis of the design process and outcomes is needed from the social justice perspective. It is also important to note that we identified certain empowerment theories featuring in the design outcomes, but we did not educate children about these theories. Instead, we in different ways nudged them towards acknowledging these theories, while it was a surprise also for us to find such a variety of them in the outcomes. Future research could explore how to discuss these theories more explicitly with children as a basis for their design. We also point out that regarding motivational empowerment, we could have analyzed the different aspects influencing it in more detail; we mostly concentrated on the meaningfulness. We also wish to underscore that as design research is about "proliferation of new realities" and is generative, i.e., changing the context in which it operates, our knowledge contribution is generative and suggestive as well [48]. In this study we have been explicating some theories underlying and embedded in the design process and outcomes [48], while this does not represent a comprehensive account of theories or other drivers shaping the design process and outcomes. Rather, this represents an account that gives a way too neat of a picture of the messy reality involved in this work. Work in schools, with teachers and children in the middle of the pandemic, is a challenging endeavor with many moving parts and changes to the plan. Even without the pandemic, we remind IS researchers to prepare for the unexpected and, if working children, for their excitement and energy as well as for their lack thereof.

In the future we wish to continue exploring and probing this messy reality of 'doing design research' and invite also other IS researchers to do so. Additionally, we invite IS researchers to tackle the crucial topics of empowerment and social justice–by the means of design of digital technology, with children as well as with other stakeholder groups.

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